

BRANCH DOCUMENT — IRIS-3

Branch: branch_20260512_010428 | Base: main-v1.0

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Decision: GO_WITH_CONSTRAINTS | Confidence: 53%

CHANGE REQUEST

delet one solarRemove the right solar panel

CASCADE ISSUES IDENTIFIED

[MAJOR] ISSUE-001 — Power generation reduced

Subsystem: solar | Req: L1-PWR-001 | Type: constraint_violation

Removing the right solar panel reduces power generation capacity, potentially violating the peak power requirement during imaging mode.

[MINOR] ISSUE-002 — Structural balance altered

Subsystem: mechanical | Req: None | Type: reanalysis_required

The removal of the right solar panel may alter the structural balance and center of mass, potentially affecting mechanical stability during launch.

[MINOR] ISSUE-003 — Thermal balance disrupted

Subsystem: thermal | Req: None | Type: reanalysis_required

The removal of the right solar panel may disrupt thermal balance, affecting temperature control and stability across orbital cycles.

MISSING ECSS TESTS — NOW REQUIRED

ECSS-E-ST-32C Rev.1 §9 — Shock test to validate structural integrity after solar panel removal

Reason: The change may affect the shock response of the payload.

ECSS-E-ST-31C §5.4 — Thermal balance test to assess thermal stability post-change

Reason: Removing a solar panel can impact thermal control and stability.

SUBSYSTEM STATUS — UPDATED

Subsystem	Score	Status	Detail	Changed
optical	92%	OK	MTF 0.42 > threshold 0.40	—
thermal	53%	WARN	Thermal balance disrupted	■ MODIFIED
mechanical	80%	OK	Structural balance altered	■ MODIFIED
structure	91%	OK	Mass 33.8 kg / budget 35 kg	—
solar	69%	WARN	Power generation reduced	■ MODIFIED
startracker	67%	WARN	Stray light analysis pending	—
dataproc	71%	WARN	Compression ratio TBC	—

AFFECTED REQUIREMENTS

- L1-PWR-001 — verification evidence invalidated, re-analysis required